

OSSP SKILL Q8:-

Detect Variable References, Expand Values, Handle Undefined Variables, Update Tokens, Support Nested Variables, Test Expansion Logic.”

CODE:-

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <string.h>
```

```
#define MAX 1000
```

```
void expand_variables(char *input)
```

```
{
```

```
    char result[MAX] = "";
```

```
    char var[100];
```

```
    int i = 0;
```

```
    while (input[i] != '\0')
```

```
    {
```

```
        if (input[i] == '$')
```

```
        {
```

```
            i++;
```

```
            int j = 0;
```

```
            // ${VARIABLE} format
```

```
            if (input[i] == '{')
```

```

{
    i++;

    while (input[i] != '}' && input[i] != '\0')
        var[j++] = input[i++];

    if (input[i] == '}')
        i++;
}
// $VARIABLE format
else
{
    while ((input[i] >= 'A' && input[i] <= 'Z') ||
           (input[i] >= 'a' && input[i] <= 'z') ||
           (input[i] >= '0' && input[i] <= '9') ||
           input[i] == '_')
    {
        var[j++] = input[i++];
    }
}

var[j] = '\0';

char *value = getenv(var);

if (value != NULL)

```

```
        strcat(result, value);
    else
        strcat(result, "");
    }
    else
    {
        int len = strlen(result);
        result[len] = input[i];
        result[len + 1] = '\0';
        i++;
    }
}
```

```
strcpy(input, result);
}
```

```
int main()
{
    char input[MAX];

    printf("Variable Expansion Shell\n");
    printf("Enter a command: ");

    fgets(input, MAX, stdin);
    input[strcspn(input, "\n")] = '\0';
```

```

printf("\nBefore expansion:\n");

printf("%s\n", input);


expand_variables(input);


printf("\nAfter expansion:\n");

printf("%s\n", input);


return 0;
}

```

OUTPUT:-

UW PICO 5.09
File: skillq8.c
Modified

```

        strcpy(input, result);
    }

int main()
{
    char input[MAX];

    printf("Variable Expansion Shell\n");
    printf("Enter a command: ");

    fgets(input, MAX, stdin);
    input[strcspn(input, "\n")] = '\0';

    printf("\nBefore expansion:\n");
    printf("%s\n", input);

    expand_variables(input);

```

```
[apple@apples-MacBook-Pro-2 skill % ./skillq8  
Variable Expansion Shell  
Enter a command: $user
```

```
Before expansion:  
$user
```

```
After expansion:
```

```
[apple@apples-MacBook-Pro-2 skill % $HOME  
zsh: permission denied: /Users/apple  
[apple@apples-MacBook-Pro-2 skill % $USER  
zsh: command not found: apple
```